SOUND

Class 9 | Science | Chapter 12

PRODUCTION & PROPAGATION OF SOUND

Sound is produced by vibrating objects, and travels as a wave that needs a MATERIAL MEDIUM (solid, liquid or gas) to travel -- it cannot travel through vacuum.

Ex: A tuning fork, vibrating vocal cords, a plucked guitar string.

Sound travels as a longitudinal wave -- particles of the medium vibrate parallel to the direction of the wave, creating compressions and rarefactions.

COMPRESSIONS & RAREFACTIONS

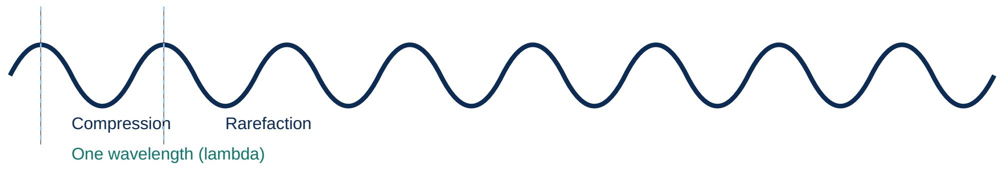
Term	Meaning
Compression	Region of high pressure & high particle density
Rarefaction	Region of low pressure & low particle density

CHARACTERISTICS OF SOUND WAVES

Wavelength, Frequency, Amplitude, Speed

DIAGRAM: A LONGITUDINAL SOUND WAVE

COMPRESSIONS AND RAREFACTIONS



KEY WAVE TERMS

Term	Meaning	Unit
Wavelength (lambda)	Distance between two consecutive compressions (or rarefactions)	metre (m)
Frequency (nu)	Number of complete oscillations (vibrations) per second	hertz (Hz)
Time period (T)	Time for one complete oscillation; T = 1/frequency	second (s)
Amplitude (A)	Maximum displacement of particles from their mean position -- decides LOUDNESS	metre (m)

Speed of sound = Wavelength x Frequency (v = lambda x nu)

SPEED OF SOUND IN DIFFERENT MEDIA

Speed: Solid > Liquid > Gas (fastest in solids, slowest in gases).

Speed of sound in air (at 25 degree C) is about 346 m/s; much faster in water and steel.

LOUDNESS, PITCH & REFLECTION

Ultrasound, Echo, SONAR

LOUDNESS & PITCH

Property	Depends on	Effect
Loudness	Amplitude of the wave	Larger amplitude = louder sound
Pitch (shrillness)	Frequency of the wave	Higher frequency = higher/shriller pitch

Quality (timbre) helps distinguish sounds of the same loudness & pitch from different sources.

AUDIBLE RANGE

Range	Frequency	Term
Human hearing	20 Hz - 20,000 Hz	Audible range
Below 20 Hz	< 20 Hz	Infrasonic sound
Above 20,000 Hz	> 20,000 Hz	Ultrasonic sound

REFLECTION OF SOUND -- ECHO

An echo is the sound heard after reflection from a surface; needs the reflecting surface to be at least 17.2 m away.

Distance = (Speed of sound x Time) / 2 -- since sound travels to the surface and back.

SONAR

Sound Navigation And Ranging -- uses ultrasonic waves to measure the distance, direction & speed of underwater objects (e.g. finding the depth of the sea).

SUMMARY & REVISION

Key Formulae + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Sound** -> produced by vibration, needs a medium to travel
- **Longitudinal wave** -> compressions & rarefactions
- **v = lambda x nu** -> speed = wavelength x frequency
- **Loudness** -> depends on amplitude
- **Pitch** -> depends on frequency
- **Echo** -> reflected sound, needs surface >=17.2 m away
- **Ultrasound** -> frequency above 20,000 Hz, used in SONAR & medical imaging

SOLVED EXAMPLE

Q: A sound wave has a frequency of 500 Hz and wavelength 0.66 m. Find its speed.

A: Using v = lambda x nu: v = 0.66 x 500 = 330 m/s.

ONE-LINE RECAP FOR REVISION

Sound needs a medium; travels fastest in solids, slowest in gases.

Loudness -> amplitude; Pitch -> frequency.

Human audible range: 20 Hz - 20,000 Hz; echo needs a distant surface.

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